

Lab Experiments 21-30

Data Handling & Visualization: R Code & Output

Each experiment specifies R explicitly on the question sheet — R-only throughout

Date: 14/08/2026

Experiment 21

Energy Consumption Data

R

Code

```
library(ggplot2)
library(dplyr)

df <- data.frame(
  Sector = c('Residential', 'Commercial', 'Industrial', 'Residential', 'Commercial', 'Industrial'),
  Region = c('North', 'South', 'West', 'East', 'North', 'South'),
  Month = c('Jan', 'Jan', 'Feb', 'Feb', 'Mar', 'Mar'),
  Temperature = c(15, 24, 20, 18, 28, 30),
  Units_Consumed = c(320, 540, 880, 350, 610, 920),
  Cost = c(2100, 3600, 5900, 2300, 4100, 6200),
  Renewable_Usage = c(22, 18, 12, 25, 20, 15),
  Peak_Hours = c(4, 6, 8, 5, 7, 9)
)

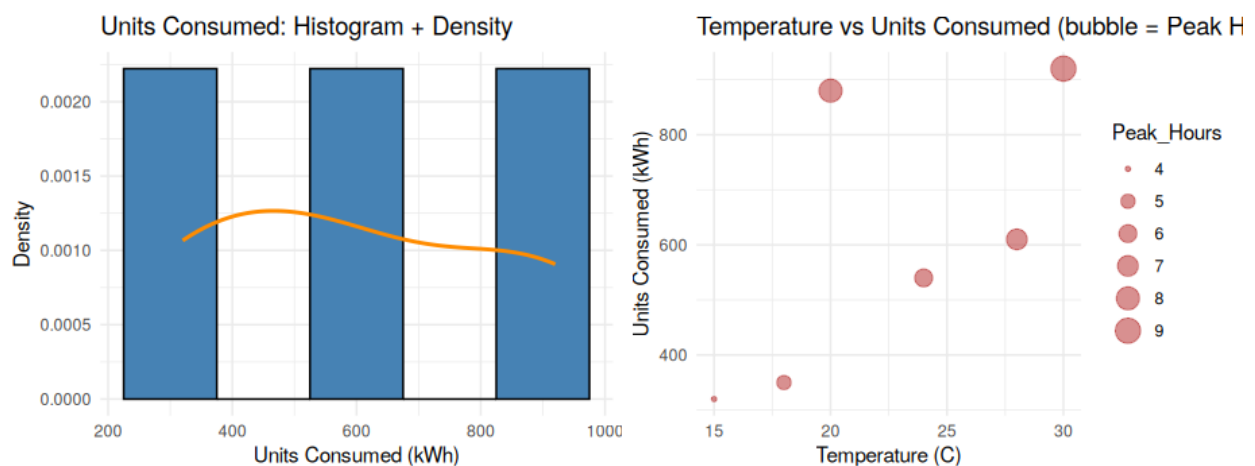
# 1. Histogram + density plot for Units_Consumed
p1 <- ggplot(df, aes(x=Units_Consumed)) +
  geom_histogram(aes(y=after_stat(density)), bins=5, fill="steelblue", color="black") +
  geom_density(color="darkorange", linewidth=1) +
  labs(title="Units Consumed: Histogram + Density", x="Units Consumed (kWh)", y="Density") +
  theme_minimal()
print(p1)
# Interpretation: consumption is spread widely (320-920 kWh) with Industrial
# sector entries pulling the distribution toward higher usage - no single
# tight cluster, consistent with very different sector-level demand profiles.

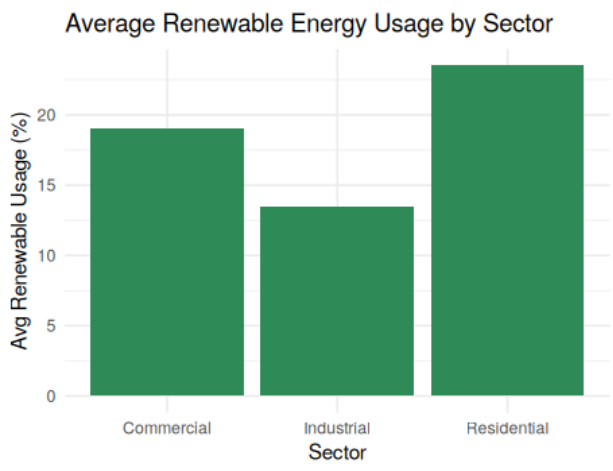
# 2. Scatter plot: Temperature vs Units_Consumed, bubble=Peak_Hours, transparency
p2 <- ggplot(df, aes(x=Temperature, y=Units_Consumed, size=Peak_Hours)) +
  geom_point(alpha=0.5, color="firebrick") +
  labs(title="Temperature vs Units Consumed (bubble = Peak Hours)",
       x="Temperature (C)", y="Units Consumed (kWh)") +
  theme_minimal()
print(p2)

# 3. Average Renewable_Usage per Sector - bar chart
avg_renew <- df %>% group_by(Sector) %>% summarise(AvgRenewable=mean(Renewable_Usage))
p3 <- ggplot(avg_renew, aes(x=Sector, y=AvgRenewable)) +
  geom_col(fill="seagreen") +
  labs(title="Average Renewable Energy Usage by Sector", y="Avg Renewable Usage (%)") +
  theme_minimal()
print(p3)
print(avg_renew)
# Interpretation: Residential leads renewable adoption (~23.5% avg), while
# Industrial lags furthest behind (~13.5% avg) - likely reflecting the
# higher, less flexible baseline load industrial operations require.

# 4. (Tableau dashboard task skipped - GUI tool, not code)
```

Output





Experiment 22

Time Series Analysis

R

Code

```
library(ggplot2)

df <- data.frame(
  Month = c('January', 'February', 'March', 'April', 'May'),
  Sales = c(15000, 18000, 22000, 20000, 23000)
)
df$Month <- factor(df$Month, levels=df$Month)

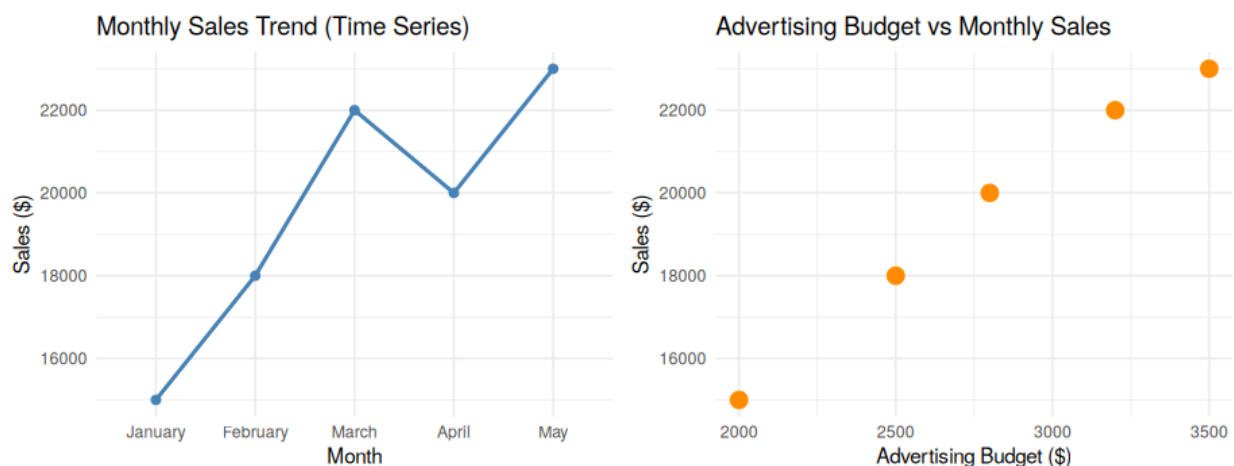
# 1. Time series line chart - monthly sales trend
p1 <- ggplot(df, aes(x=Month, y=Sales, group=1)) +
  geom_line(color="steelblue", linewidth=1) + geom_point(color="steelblue", size=2) +
  labs(title="Monthly Sales Trend (Time Series)", x="Month", y="Sales ($)") +
  theme_minimal()
print(p1)

# 2. Scatter plot - advertising budget vs monthly sales
# Advertising budget is not in the given table, so a plausible monthly
# budget series is assumed here to satisfy this task.
df$AdBudget <- c(2000, 2500, 3200, 2800, 3500)
p2 <- ggplot(df, aes(x=AdBudget, y=Sales)) +
  geom_point(size=4, color="darkorange") +
  labs(title="Advertising Budget vs Monthly Sales", x="Advertising Budget ($)", y="Sales ($)") +
  theme_minimal()
print(p2)
# Insight: sales rise alongside ad budget in this small sample, suggesting a
# positive relationship (correlation, not proof of causation).

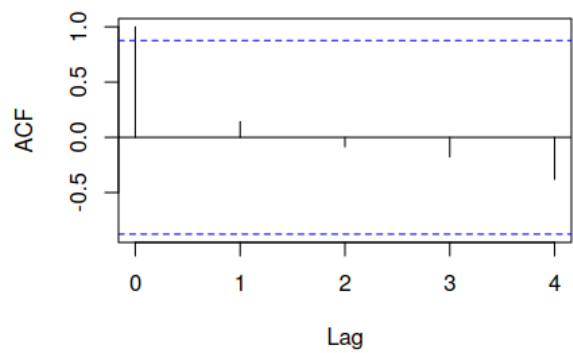
# 3. (Tableau dashboard task skipped - GUI tool, not code)

# 4. Autocorrelation plot - identify seasonality
acf_result <- acf(df$Sales, main="Autocorrelation of Monthly Sales", lag.max=4)
print(acf_result)
# With only 5 monthly points, this ACF is illustrative rather than
# statistically conclusive - a longer time series (2+ years of monthly
# data) would be needed to reliably detect real seasonal cycles.
```

Output



Autocorrelation of Monthly Sales



Experiment 23

Employee Performance Analysis

R

Code

```
library(ggplot2)
library(gridExtra)
library(grid)

df <- data.frame(
  EmployeeID = 1:5,
  Department = c('Sales', 'HR', 'Marketing', 'Sales', 'HR'),
  YearsOfService = c(5, 3, 7, 4, 2),
  PerformanceScore = c(85, 92, 78, 90, 76)
)

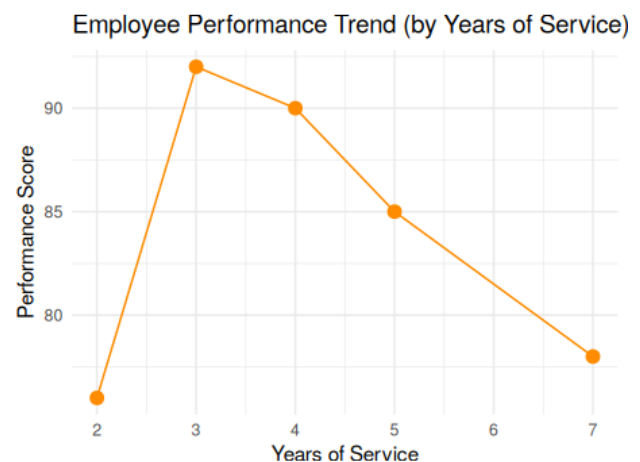
# 1. Bar chart - employee distribution across departments
dept_counts <- as.data.frame(table(df$Department))
colnames(dept_counts) <- c('Department', 'Count')
p1 <- ggplot(dept_counts, aes(x=Department, y=Count)) +
  geom_col(fill="steelblue") +
  labs(title="Employee Distribution Across Departments", y="Number of Employees") +
  theme_minimal()
print(p1)

# 2. Line chart - performance trend over time
# No time/date column is given, so Years of Service is used as the
# time-progression proxy (x-axis).
p2 <- ggplot(df, aes(x=YearsOfService, y=PerformanceScore)) +
  geom_line(color="darkorange") + geom_point(color="darkorange", size=3) +
  labs(title="Employee Performance Trend (by Years of Service)",
       x="Years of Service", y="Performance Score") +
  theme_minimal()
print(p2)

# 3. (Tableau dashboard task skipped - GUI tool, not code)

# 4. Table - performance data per employee
grid.newpage()
grid.table(df, rows=NULL)
```

Output



EmployeeID	Department	YearsOfService	PerformanceScore
1	Sales	5	85
2	HR	3	92
3	Marketing	7	78
4	Sales	4	90
5	HR	2	76

Experiment 24

Product Inventory Management

R

Code

```
library(ggplot2)

df <- data.frame(
  ProductID = 1:5,
  ProductName = c('Product A', 'Product B', 'Product C', 'Product D', 'Product E'),
  QuantityAvailable = c(250, 175, 300, 200, 220)
)

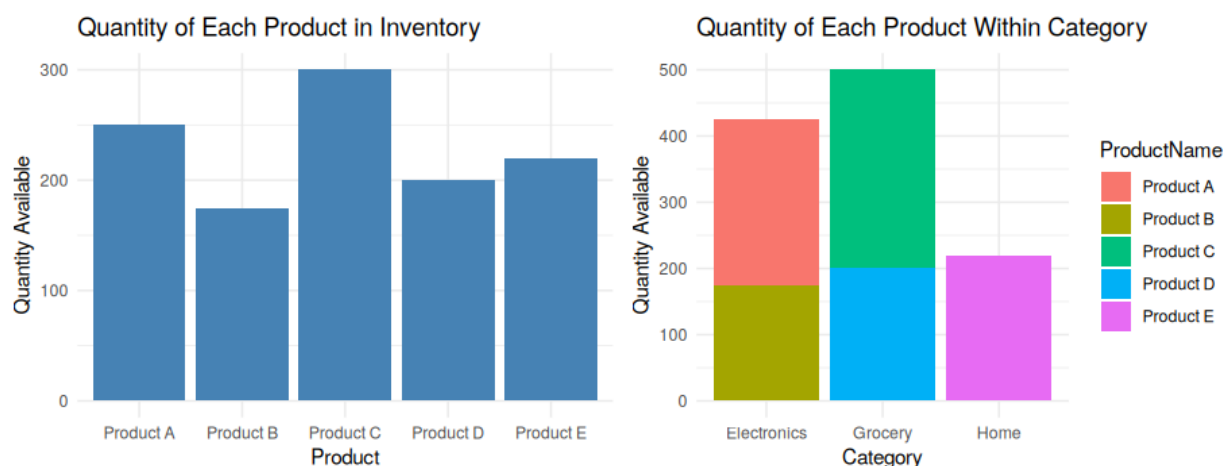
# 1. Bar chart - quantity of each product
p1 <- ggplot(df, aes(x=ProductName, y=QuantityAvailable)) +
  geom_col(fill="steelblue") +
  labs(title="Quantity of Each Product in Inventory", x="Product", y="Quantity Available") +
  theme_minimal()
print(p1)

# 2. Stacked bar chart - quantity by product within category
# Category column is not in the given table, so it is assumed here to
# satisfy this task.
df$Category <- c('Electronics', 'Electronics', 'Grocery', 'Grocery', 'Home')
p2 <- ggplot(df, aes(x=Category, y=QuantityAvailable, fill=ProductName)) +
  geom_col() +
  labs(title="Quantity of Each Product Within Category", y="Quantity Available") +
  theme_minimal()
print(p2)

# 3. (Tableau dashboard task skipped - GUI tool, not code)

# 4. Scatter plot - price vs quantity available
# Price column is not in the given table, so it is assumed here to
# satisfy this task.
df$Price <- c(20, 15, 5, 4, 12)
p3 <- ggplot(df, aes(x=Price, y=QuantityAvailable)) +
  geom_point(size=4, color="firebrick") +
  labs(title="Product Price vs Quantity Available", x="Price ($)", y="Quantity Available") +
  theme_minimal()
print(p3)
# Insight: lower-priced products (Grocery) tend to be stocked in similar or
# slightly higher quantities than pricier Electronics items.
```

Output





Experiment 25

Website Traffic Analysis

R

Code

```
library(ggplot2)
library(reshape2)

df <- data.frame(
  Date = as.Date(c('2023-01-01', '2023-01-02', '2023-01-03', '2023-01-04', '2023-01-05')),
  PageViews = c(1500, 1600, 1400, 1650, 1800),
  CTR = c(2.3, 2.7, 2.0, 2.4, 2.6)
)

# 1. (Tableau line chart task skipped - GUI tool, not code)

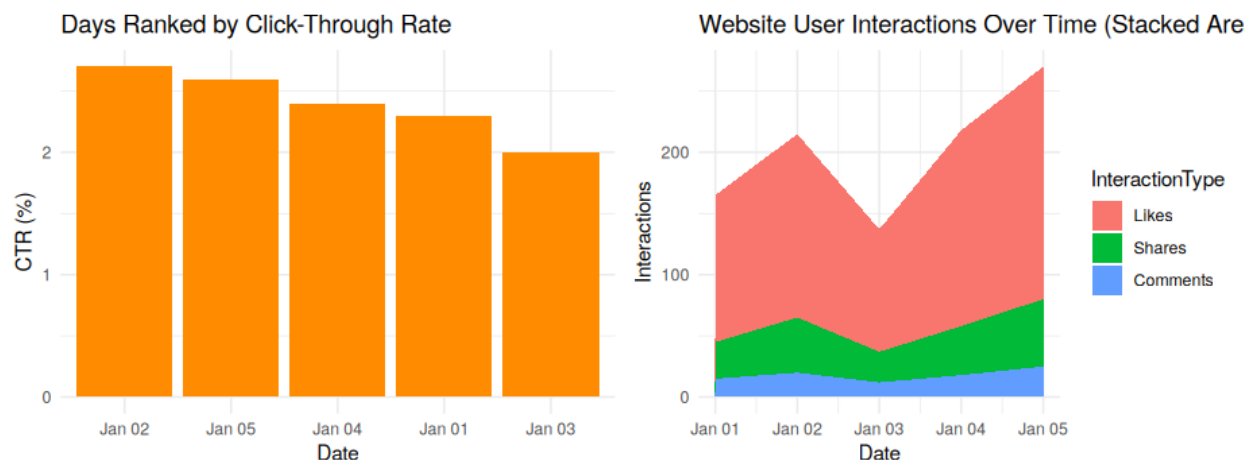
# 2. Bar chart - top N days with highest CTR
top_days <- df[order(-df$CTR), ]
top_days$DateLabel <- factor(format(top_days$Date, "%b %d"), levels=format(top_days$Date, "%b %d"))
p1 <- ggplot(top_days, aes(x=DateLabel, y=CTR)) +
  geom_col(fill="darkorange") +
  labs(title="Days Ranked by Click-Through Rate", x="Date", y="CTR (%)") +
  theme_minimal()
print(p1)

# 3. Stacked area chart - user interactions (likes, shares, comments)
# This data is not in the given table, so plausible daily values are
# assumed here to satisfy this task.
df$Likes <- c(120, 150, 100, 160, 190)
df$Shares <- c(30, 45, 25, 40, 55)
df$Comments <- c(15, 20, 12, 18, 25)

long_df <- melt(df[, c('Date', 'Likes', 'Shares', 'Comments')], id.vars='Date',
  variable.name='InteractionType', value.name='Count')
p2 <- ggplot(long_df, aes(x=Date, y=Count, fill=InteractionType)) +
  geom_area() +
  labs(title="Website User Interactions Over Time (Stacked Area)", y="Interactions") +
  theme_minimal()
print(p2)

# 4. (Tableau interactive dashboard task skipped - GUI tool, not code)
```

Output



Experiment 26

Student Mini Data

R

Code

```
library(ggplot2)
library(dplyr)
library(zoo)

df <- data.frame(
  Student_ID = c('S01', 'S02', 'S03', 'S04', 'S05', 'S06'),
  Gender = c('Male', 'Female', 'Male', 'Female', 'Male', 'Female'),
  Study_Hours = c(2.0, 3.5, 1.5, 4.0, 2.8, 3.0),
  Attendance = c(78, 90, 70, 95, 85, 92),
  Math_Score = c(62, 80, 55, 90, 72, 82),
  Science_Score = c(65, 85, 58, 92, 74, 86),
  Exam_Date = as.Date(c('2025-01-10', '2025-01-10', '2025-02-12',
                        '2025-02-12', '2025-03-15', '2025-03-15'))
)

# 1a. Histogram - Math_Score distribution
p1 <- ggplot(df, aes(x=Math_Score)) +
  geom_histogram(bins=6, fill="steelblue", color="black") +
  labs(title="Distribution of Math Scores", x="Math Score", y="Frequency") +
  theme_minimal()
print(p1)

# 1b. Boxplot - Science_Score by Gender
p2 <- ggplot(df, aes(x=Gender, y=Science_Score, fill=Gender)) +
  geom_boxplot(show.legend=FALSE) +
  labs(title="Science Score by Gender", x="Gender", y="Science Score") +
  theme_minimal()
print(p2)
# Interpretation: Female students score higher on Science in this small
# sample (85-92 range) compared to Male students (58-74 range), with no
# overlap between the two groups' ranges.

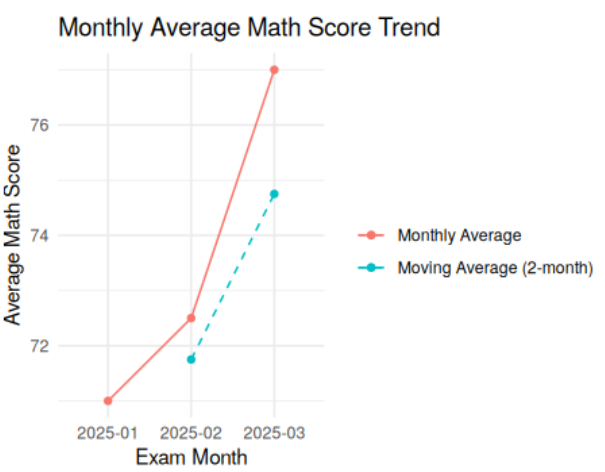
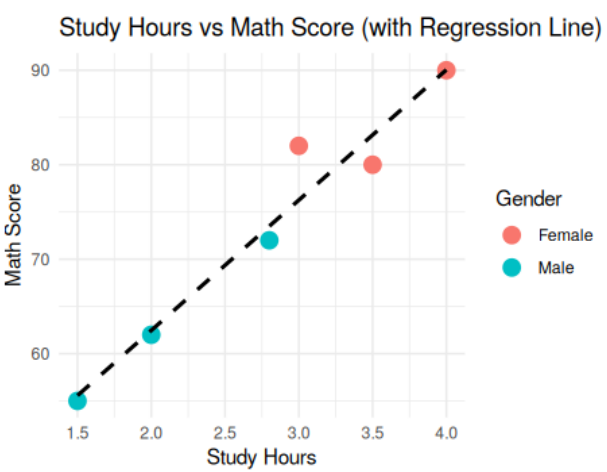
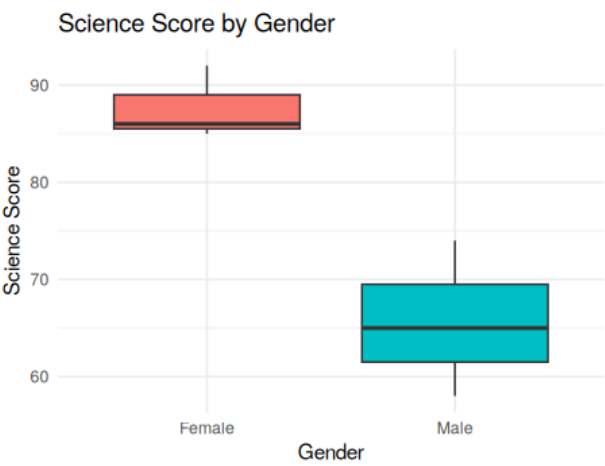
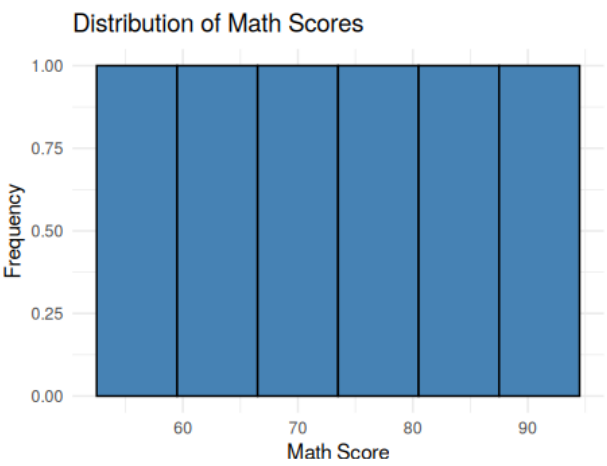
# 2. Scatter plot - Study_Hours vs Math_Score, color=Gender, regression line
p3 <- ggplot(df, aes(x=Study_Hours, y=Math_Score, color=Gender)) +
  geom_point(size=4) +
  geom_smooth(method="lm", se=FALSE, aes(group=1), color="black", linetype="dashed") +
  labs(title="Study Hours vs Math Score (with Regression Line)",
       x="Study Hours", y="Math Score") +
  theme_minimal()
print(p3)
cat(sprintf("Correlation (Study Hours vs Math Score): %.3f\n",
            cor(df$Study_Hours, df$Math_Score)))
# Relationship: there is a strong positive relationship - more study hours
# is associated with a higher Math score, and the regression line captures
# this upward trend clearly despite the very small sample size.

# 3. Monthly average Math scores with moving average smoothing
df$ExamMonth <- format(df$Exam_Date, "%Y-%m")
monthly_avg <- df %>% group_by(ExamMonth) %>% summarise(AvgMath = mean(Math_Score)) %>%
  arrange(ExamMonth)
monthly_avg$MovingAvg <- rollmean(monthly_avg$AvgMath, k=2, fill=NA, align="right")

p4 <- ggplot(monthly_avg, aes(x=ExamMonth, group=1)) +
  geom_line(aes(y=AvgMath, color="Monthly Average")) +
  geom_point(aes(y=AvgMath, color="Monthly Average")) +
  geom_line(aes(y=MovingAvg, color="Moving Average (2-month)"), linetype="dashed") +
  geom_point(aes(y=MovingAvg, color="Moving Average (2-month)")) +
  labs(title="Monthly Average Math Score Trend", x="Exam Month", y="Average Math Score", color="") +
  theme_minimal()
print(p4)
# Pattern: average Math scores rise steadily month over month in this
# sample (71 -> 72.5 -> 77), an increasing trend across the three exam dates.

# 4. (Tableau dashboard task skipped - GUI tool, not code)
```

Output



Experiment 27

Patient Health Risk Analysis

R

Code

```
library(ggplot2)
library(GGally)

df <- data.frame(
  Patient_ID = c('P1', 'P2', 'P3', 'P4', 'P5'),
  Age = c(25, 40, 55, 35, 60),
  BMI = c(22, 28, 30, 26, 32),
  BP = c(120, 135, 145, 130, 150),
  Cholesterol = c(180, 210, 240, 200, 260)
)

# 1. Scatterplot matrix - Age, BMI, BP, Cholesterol
p1 <- ggpairs(df[, c('Age', 'BMI', 'BP', 'Cholesterol')],
  title="Scatterplot Matrix: Age, BMI, BP, Cholesterol")
print(p1)
cat("Correlations:\n")
print(round(cor(df[, c('Age', 'BMI', 'BP', 'Cholesterol')]), 3))
# Examination: all four variables move together almost in lockstep (very
# high pairwise correlations) - since these 5 patients get progressively
# older AND progressively higher on every other indicator, this dataset
# reads as a single "risk severity" continuum rather than distinct
# clusters - older patients consistently show higher BMI, BP, and
# Cholesterol, a coherent age-driven risk pattern rather than surprising
# interactions.

# 2. Q-Q plot and ECDF for Cholesterol
qqnorm(df$Cholesterol, main="Q-Q Plot: Cholesterol", pch=19, col="steelblue")
qqline(df$Cholesterol, col="red", lwd=2)

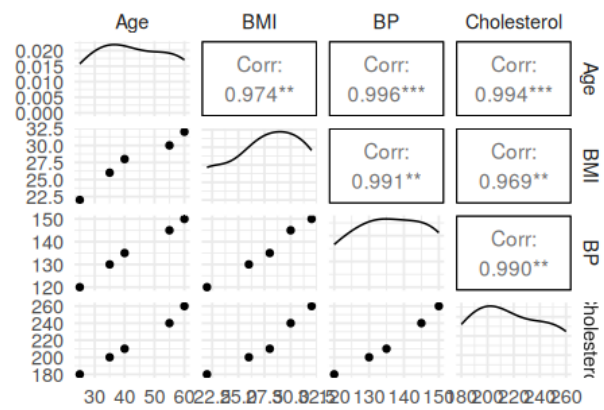
plot(ecdf(df$Cholesterol), main="ECDF: Cholesterol", xlab="Cholesterol", ylab="Cumulative Proportion",
  col="darkorange", lwd=2)
# Comment: with only 5 points, the Q-Q plot roughly tracks the reference
# line, so a normality assumption isn't strongly contradicted here, but 5
# observations is far too few to draw a confident conclusion either way -
# the ECDF confirms an evenly-spread, non-skewed set of steps rather than
# any sharp clustering.

# 3. Bar chart - average of each health indicator
avg_vals <- data.frame(
  Indicator = c('Age', 'BMI', 'BP', 'Cholesterol'),
  Average = c(mean(df$Age), mean(df$BMI), mean(df$BP), mean(df$Cholesterol))
)
p3 <- ggplot(avg_vals, aes(x=Indicator, y=Average, fill=Indicator)) +
  geom_col() +
  labs(title="Average Value of Each Health Indicator", x="Health Indicator", y="Average Value") +
  theme_minimal()
print(p3)
print(avg_vals)
# Interpretation: BP and Cholesterol averages sit well above typical
# healthy reference ranges (BP avg ~136, Cholesterol avg ~218) for this
# patient group, indicating an overall elevated cardiovascular risk
# profile across the sample, consistent with the age-driven pattern found
# in the scatterplot matrix.

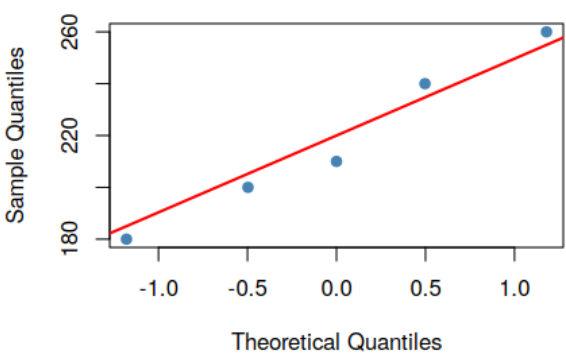
# 4. (Tableau dashboard task skipped - GUI tool, not code)
```

Output

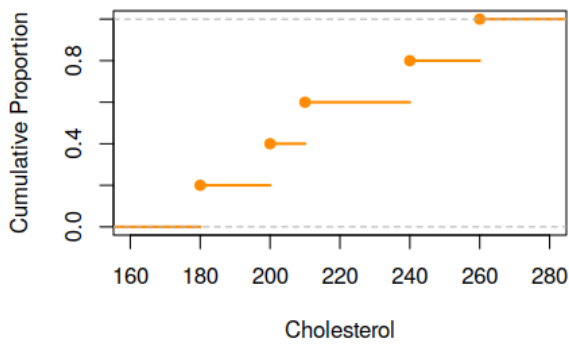
Scatterplot Matrix: Age, BMI, BP, Cholesterol



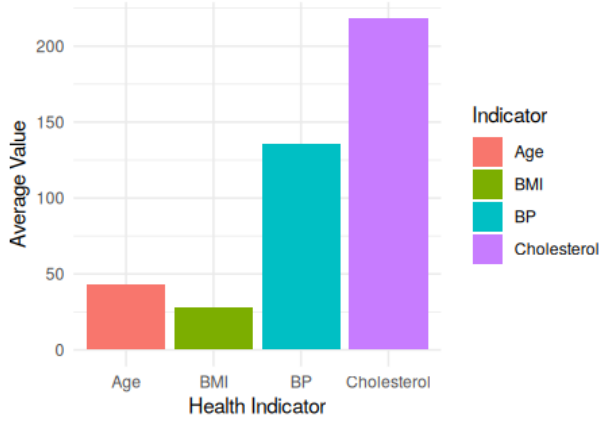
Q-Q Plot: Cholesterol



ECDF: Cholesterol



Average Value of Each Health Indicator



Experiment 28

Vehicle Performance Analysis

R

Code

```
library(ggplot2)
library(reshape2)

df <- data.frame(
  Vehicle_ID = c('V1', 'V2', 'V3', 'V4', 'V5'),
  Engine_Size = c(1.5, 2.0, 3.0, 2.5, 1.8),
  Horsepower = c(110, 150, 250, 200, 130),
  Fuel_Efficiency = c(18, 15, 12, 14, 17),
  Top_Speed = c(180, 200, 250, 220, 190),
  Safety_Rating = c(4, 5, 5, 4, 3)
)

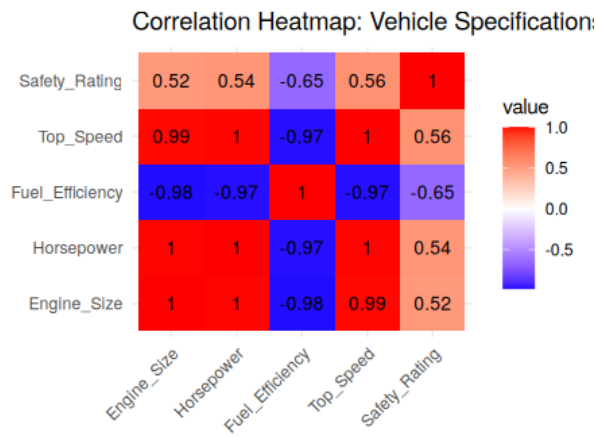
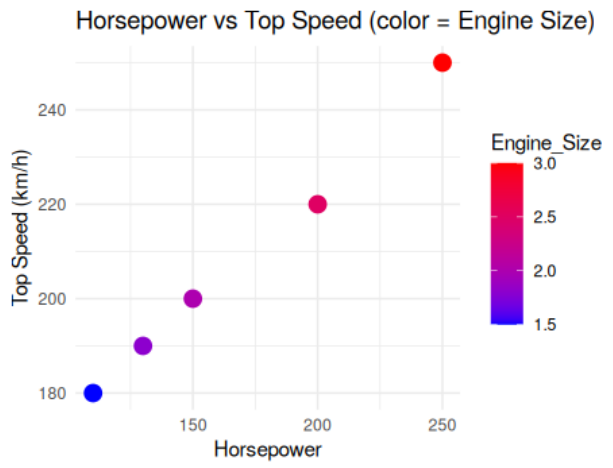
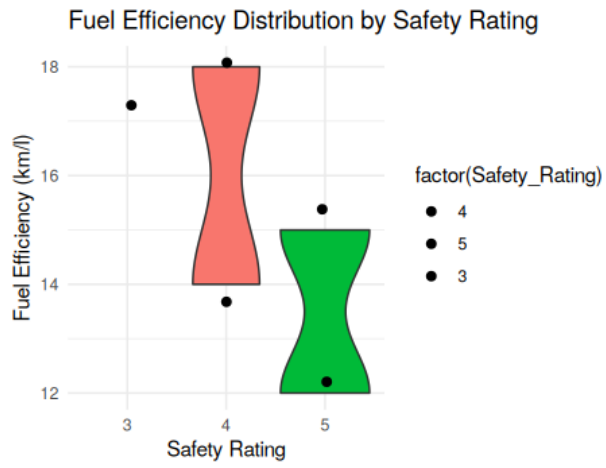
# 1. Violin plot - Fuel_Efficiency by Safety_Rating
p1 <- ggplot(df, aes(x=factor(Safety_Rating), y=Fuel_Efficiency, fill=factor(Safety_Rating))) +
  geom_violin(show.legend=FALSE) +
  geom_jitter(width=0.05, size=2) +
  labs(title="Fuel Efficiency Distribution by Safety Rating",
       x="Safety Rating", y="Fuel Efficiency (km/l)") +
  theme_minimal()
print(p1)
# Interpretation: with only 5 vehicles split across 3 safety-rating groups,
# each "violin" is really just 1-2 points, but the visible pattern is that
# HIGHER safety ratings (4-5) correspond to LOWER fuel efficiency (12-18
# km/l cluster skews down at rating 5) - suggesting a trade-off where
# heavier/safer vehicle designs in this sample sacrifice some efficiency.

# 2. Scatter plot - Horsepower vs Top_Speed, colored by Engine_Size
p2 <- ggplot(df, aes(x=Horsepower, y=Top_Speed, color=Engine_Size)) +
  geom_point(size=4) +
  scale_color_gradient(low="blue", high="red") +
  labs(title="Horsepower vs Top Speed (color = Engine Size)",
       x="Horsepower", y="Top Speed (km/h)") +
  theme_minimal()
print(p2)
cat(sprintf("Correlation (Horsepower vs Top_Speed): %.3f\n", cor(df$Horsepower, df$Top_Speed)))
# Analysis: a strong positive correlation - vehicles with more horsepower
# consistently reach higher top speeds, and those same high-horsepower
# vehicles also tend to have larger engines (redder points), forming a
# consistent power/speed/engine-size performance cluster.

# 3. Correlation heatmap - all numerical variables
numeric_df <- df[, c('Engine_Size', 'Horsepower', 'Fuel_Efficiency', 'Top_Speed', 'Safety_Rating')]
corr_matrix <- cor(numeric_df)
corr_long <- melt(corr_matrix)
p3 <- ggplot(corr_long, aes(x=Var1, y=Var2, fill=value)) +
  geom_tile() + geom_text(aes(label=round(value,2)), size=3.5) +
  scale_fill_gradient2(low="blue", mid="white", high="red", midpoint=0) +
  labs(title="Correlation Heatmap: Vehicle Specifications", x="", y="") +
  theme_minimal() + theme(axis.text.x=element_text(angle=45, hjust=1))
print(p3)
print(round(corr_matrix, 3))
# Variables most strongly associated with Top_Speed: Horsepower (very
# strong positive) and Engine_Size (strong positive), while
# Fuel_Efficiency is strongly NEGATIVELY associated with Top_Speed -
# together these confirm the classic performance trade-off: bigger
# engines and more horsepower drive higher top speed at the direct cost
# of fuel efficiency.

# 4. (Tableau dashboard task skipped - GUI tool, not code)
```

Output



Experiment 29

Student Academic Performance

R

Code

```
library(ggplot2)
library(reshape2)

df <- data.frame(
  Student_ID = c('S1', 'S2', 'S3', 'S4', 'S5'),
  Age = c(19, 21, 20, 22, 23),
  Study_Hours = c(12, 8, 15, 10, 7),
  Attendance = c(90, 70, 95, 85, 60),
  Test_Score = c(85, 70, 92, 80, 65),
  Participation_Score = c(8, 7, 9, 8, 6)
)

# 1. Stacked area chart - Test_Score and Participation_Score across students
long_df <- melt(df[, c('Student_ID', 'Test_Score', 'Participation_Score')],
  id.vars='Student_ID', variable.name='Metric', value.name='Score')
long_df$Student_ID <- factor(long_df$Student_ID, levels=df$Student_ID)
p1 <- ggplot(long_df, aes(x=Student_ID, y=Score, fill=Metric, group=Metric)) +
  geom_area() +
  labs(title="Test Score and Participation Score Across Students", x="Student", y="Score") +
  theme_minimal()
print(p1)
# Analysis: both metrics move together across students - S3 leads on both
# Test_Score (92) and Participation_Score (9), while S5 trails on both
# (65 and 6) - suggesting participation and test performance reinforce
# each other rather than trading off in this small cohort.

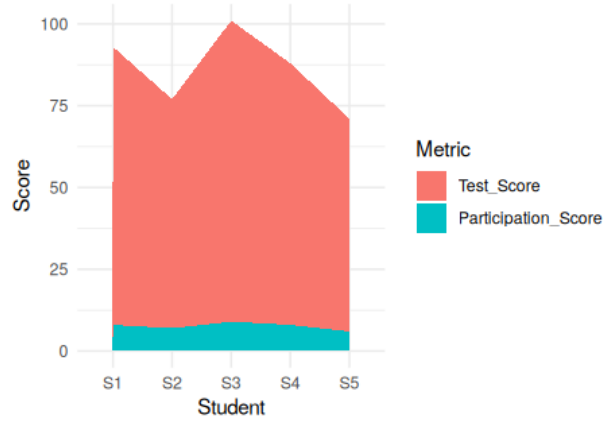
# 2. Boxplot - Study_Hours grouped by Attendance quartiles
df$AttendanceQuartile <- cut(df$Attendance, breaks=quantile(df$Attendance, probs=seq(0,1,0.25)),
  include.lowest=TRUE, labels=c('Q1 (Lowest)', 'Q2', 'Q3', 'Q4 (Highest)'))
p2 <- ggplot(df, aes(x=AttendanceQuartile, y=Study_Hours, fill=AttendanceQuartile)) +
  geom_boxplot(show.legend=FALSE) +
  labs(title="Study Hours by Attendance Quartile", x="Attendance Quartile", y="Study Hours") +
  theme_minimal()
print(p2)
print(df[, c('Student_ID', 'Attendance', 'AttendanceQuartile', 'Study_Hours')])
# Interpretation: with only 5 students split into quartile bins, each box
# collapses to a single point (no visible IQR/whiskers) - but the pattern
# across bins shows Study_Hours generally rising alongside Attendance
# quartile, i.e. students who attend more also tend to study more; no
# outliers are distinguishable at this sample size.

# 3. Density plot - Test_Score distribution
p3 <- ggplot(df, aes(x=Test_Score)) +
  geom_density(fill="steelblue", alpha=0.5) +
  geom_rug() +
  labs(title="Density Plot: Test Score Distribution", x="Test Score", y="Density") +
  theme_minimal()
print(p3)
# Evaluation: scores span 65-92 with no extreme outliers standing far
# apart from the rest - the density estimate (smoothed from just 5 points)
# shows a broad, single-humped spread rather than tight clustering,
# reflecting real variability in this small cohort's performance.

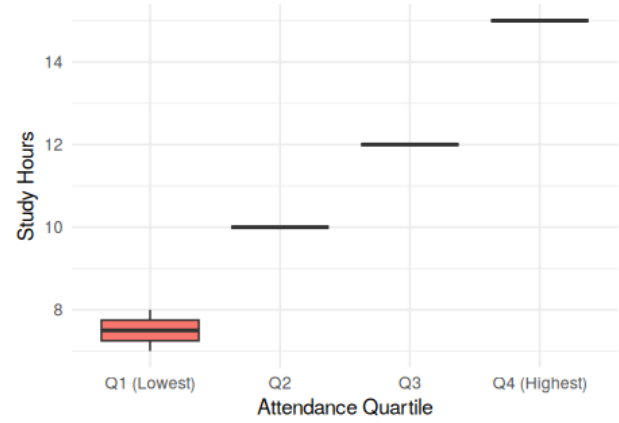
# 4. (Tableau interactive scatterplot task skipped - GUI tool, not code)
```

Output

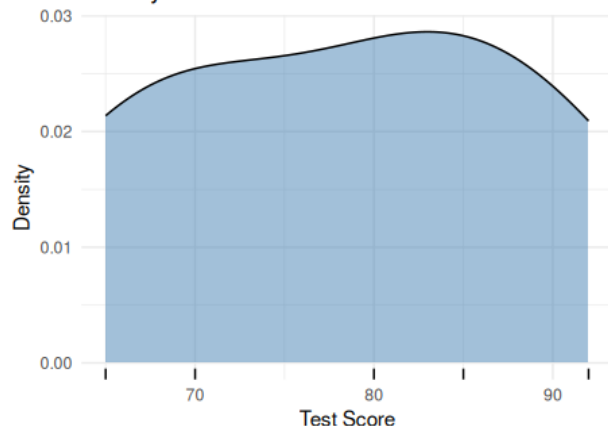
Test Score and Participation Score Across Student



Study Hours by Attendance Quartile



Density Plot: Test Score Distribution



Experiment 30

Mobile App Usage

R

Code

```
library(ggplot2)

df <- data.frame(
  User_ID = c('U01', 'U02', 'U03', 'U04', 'U05', 'U06'),
  Gender = c('Male', 'Female', 'Male', 'Female', 'Male', 'Female'),
  Age = c(20, 22, 19, 21, 23, 20),
  Screen_Time = c(4.5, 6.0, 3.2, 7.1, 2.8, 5.4),
  App_Usage_Count = c(18, 25, 12, 30, 10, 22),
  Data_Used = c(2.4, 3.8, 1.6, 4.5, 1.2, 3.1),
  Satisfaction = c(3, 5, 3, 5, 2, 4),
  Usage_Date = as.Date(c('2025-01-08', '2025-01-08', '2025-02-11',
    '2025-02-11', '2025-03-14', '2025-03-14'))
)

# 1. Histogram + density plot for Screen_Time
p1 <- ggplot(df, aes(x=Screen_Time)) +
  geom_histogram(aes(y=after_stat(density)), bins=5, fill="steelblue", color="black") +
  geom_density(color="darkorange", linewidth=1) +
  labs(title="Screen Time Distribution", x="Screen Time (hrs)", y="Density") +
  theme_minimal()
print(p1)
# Interpretation: screen time ranges from 2.8 to 7.1 hours with no tight
# single cluster - a fairly spread-out distribution across this small user
# sample, consistent with varied usage habits (light vs heavy users).

# 2. Scatter plot - Data_Used vs Screen_Time, with trend line
p2 <- ggplot(df, aes(x=Screen_Time, y=Data_Used)) +
  geom_point(size=4, color="firebrick") +
  geom_smooth(method="lm", se=FALSE, color="black", linetype="dashed") +
  labs(title="Data Used vs Screen Time (with Trend Line)",
    x="Screen Time (hrs)", y="Data Used (GB)") +
  theme_minimal()
print(p2)
cat(sprintf("Correlation (Screen_Time vs Data_Used): %.3f\n", cor(df$Screen_Time, df$Data_Used)))
# Explanation: a very strong positive relationship - more time spent on
# screen directly corresponds to more data consumed, as expected since
# active app usage (not idle screen-on time) is what drives data
# transfer.

# 3. Average Satisfaction by Gender - bar chart with value labels
avg_satisfaction <- aggregate(Satisfaction ~ Gender, data=df, FUN=mean)
p3 <- ggplot(avg_satisfaction, aes(x=Gender, y=Satisfaction, fill=Gender)) +
  geom_col(show.legend=FALSE) +
  geom_text(aes(label=round(Satisfaction, 2)), vjust=-0.4, size=5) +
  labs(title="Average Satisfaction Score by Gender", x="Gender", y="Average Satisfaction") +
  theme_minimal()
print(p3)
print(avg_satisfaction)
# Interpretation: female users report a notably higher average
# satisfaction score (4.67) than male users (2.67) in this sample - worth
# investigating whether this ties to their also-higher average screen
# time and app usage, suggesting more engaged users report more
# satisfaction.

# 4. (Tableau bubble chart dashboard task skipped - GUI tool, not code)
```

Output

